Operator-Entropy Joint-Embedding Predictive Architectures for Multivariate Time-Series Anomaly Detection

Andre Williams, Imadeldin Mahgoub, *Senior Member, IEEE,* and Hamid Akbarian

***Abstract*—Deep reconstructive detectors for multivariate time-series anomaly detection minimize observation-space error, fitting stochastic sensor noise that produces heavy-tailed residuals and severe false-alarm flooding under automated threshold calibration. We investigate Joint-Embedding Predictive Architectures (JEPAs) scoring predictive discrepancies in a low-dimensional latent space, alongside an Operator-Entropy variant (OpEntropy JEPA) whose linear Koopman operator incorporates spectral entropy and a shifted logarithmic Bregman barrier, coordinate saliency gating, and a multiscale dyadic frequency front-end. Across 33 continuous telemetry streams from seven physical datasets under chronological held-out Streaming Peaks-Over-Threshold calibration, latent predictive scoring improves relative false-alarm control three- to four-fold under retrospective buffered attribution (7.4% to 10.1% nominal false-positive rate versus 30.0% for TimesNet and 30.7% for TranAD) and eight- to twenty-fold under strictly causal streaming (1.1% to 2.8% versus 22.2% to 24.3%). This false-alarm suppression does not establish general detection superiority: point F1, precision-recall ranking, and event recall remain comparable to reconstructive baselines. Controlled mechanism ablations isolate two drivers of residual conditioning: low-dimensional LayerNorm bottlenecks compress residual kurtosis, while learned latent projections reduce inter-block distribution drift relative to normalized observation forecasters. On a fixed Koopman backbone, factorial and Two One-Sided Tests ablations confirm that spectral entropy and barrier terms preserve full operator rank without altering detection within a 0.03 F1 margin, whereas cross-architecture F1 differences against unconstrained TS-JEPA remain inconclusive despite shorter detection latency. Static extreme-value calibration still overshoots its 0.1% design risk under deployment shift; a three-stage permutation decomposition attributes 57% to 64% of excess causal overshoot in latent models to inter-block drift and 29% to 35% to serial dependence.**

***Index Terms*—Time-series anomaly detection, Multivariate analysis, Threshold calibration, Extreme value theory, Representation learning.**

I. Introduction

Autonomous anomaly detection in multi-sensor telemetry monitors operational health across aerospace systems, Earth observation, cyber-physical infrastructure, and wearable digital healthcare [1], [2]. In complex deployments (NASA SMAP, MSL rovers, industrial IoT, and biometric arrays), dozens to hundreds of channels concurrently track power buses, thermal gradients, vibrations, and kinematics [3], where faults must be identified without overwhelming operators with nuisance alerts.

False-Alarm Flooding Under Automated Threshold Calibration: Operational monitoring requires automated decision rules that convert continuous anomaly scores into binary alarms using nominal calibration data alone. Prevailing deep detectors (TimesNet, TranAD, Anomaly Transformer) minimize point-level reconstruction error in raw observation space [4]–[6]. Because raw telemetry contains ambient sensor jitter and channel-scale disparities, observation-space residuals exhibit severe excess kurtosis ($\gamma_2 > 8$) and serial clustering. When paired with automated extreme-value thresholding such as Streaming Peaks-Over-Threshold (SPOT) [7] at nominal design risk $q = 10^{-3}$ (0.1%), these heavy-tailed residuals suffer severe calibration breakdown on unseen nominal data, triggering false alarms across 22%–31% of nominal timesteps.

Latent Residual Conditioning and Scope of Claims: To suppress false-alarm flooding without synthetic anomaly injection [8], we study Joint-Embedding Predictive Architectures (JEPAs) that score predictive discrepancies in a low-dimensional latent space [9], alongside an *Operator-Entropy JEPA* (OpEntropy JEPA) variant regularized by Koopman spectral entropy and a shifted Bregman energy barrier. Physics-inspired regularizers (operator entropy, Reynolds-stress closure, curl-free potential flow) are evaluated strictly as algebraic priors on abstract latent coordinates, not as physical conservation laws, and our experiments separate three distinct operational objectives: (i) suppressing nuisance false alarms under automated calibration, (ii) detecting anomalies reliably across operating points, and (iii) emitting scores early enough to support intervention.

A. Contributions

This study makes four focused contributions:

1) **Mechanism Isolation (Compression vs. Drift):** Across 12 controlled backbone arms on all 19 non-GHL streams (Section VI-E, Fig. 2(a), Table I(e)), adding a D=32 LayerNorm bottleneck to an observation forecaster (`TCN-ObsPred-LN`) compresses calibration kurtosis (γ_2 : $6.10 \rightarrow 0.91$ causal) and halves causal FPR ($0.247 \rightarrow 0.117$), while learned latent projection in TS-JEPA and OpEntropy JEPA halves calibration-to-holdout

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Kolmogorov–Smirnov drift ($0.376 \rightarrow 0.214$, p=0.0010; $0.376 \rightarrow 0.209$, p=0.0004) and lowers causal FPR to 0.009–0.028.

2) **Relative False-Alarm Suppression vs. Detection:** Across 33 telemetry streams from seven physical datasets, latent predictive scoring improves relative held-out false-alarm control three- to four-fold buffered (7.4%–10.1% vs. 30.0%–30.7%, $p \leq 0.006$) and $8 \times$ –$20\times$ causally (1.1%–2.8% vs. 22.2%–24.3%). Separating deployable calibration quantiles (q^*) from post-hoc oracle matched realized test-FPR curves (f^*, Fig. 2(b)) shows that latent models improve low-FPR causal event recall (0.395–0.431 vs. 0.217 at f^*=0.01), whereas Point-F1, PR-AUC, and buffered recall remain comparable to reconstructive baselines.
3) **Operator-Entropy Role and Equivalence Bounds:** Proposition 1 proves that at finite $\epsilon > 0$, $\nabla_{\mathbf{K}} \mathcal{R}_{\text{spectral}}(\mathbf{0}) = \mathbf{0}$ and non-zero cross-covariance ($\mathbf{M} \neq \mathbf{0}$) excludes zero stationarity, whereas as $\epsilon \rightarrow 0^+$ the barrier diverges as $\|\mathbf{K}\|_F \rightarrow 0$. The OpEntropy architecture (Koopman $\mathbf{K}$, detached Lyapunov alignment, relative radial energy score) cuts buffered delay from 110.3 to 79.5 steps ($127.0 \rightarrow 58.2$ causal). On the fixed Koopman backbone (λ_{ent}=0 control), Two One-Sided Tests (TOST) confirm equivalence within $\Delta = \pm 0.03$ F1 (90% CI $[-0.011, +0.003]$, $p_{\text{TOST}} = 0.00074$), while cross-architecture F1 differences vs. TS-JEPA are inconclusive on the 19-stream subset ($+0.028$, 90% CI $[-0.018, +0.074]$, $p_{\text{TOST}} = 0.11$) and bounded within ± 0.05 across all 33 streams ($+0.016$, $p_{\text{TOST}} = 0.0019$).
4) **Calibration Breakdown and Drift Decomposition:** Static thresholding misses its nominal q=10^{-3} target out of sample (1.5%–2.0% stream-macro causal and 5.5%–6.4% buffered FPR across 33 streams for JEPAs vs. 14%–20% for reconstructors, Fig. 2(c)). A three-stage permutation protocol (chronological $\rightarrow$ block-64 $\rightarrow$ pointwise i.i.d.) separates inter-block shift (56.6%–64.1% of excess causal overshoot for JEPAs; 77.5%–92.3% for reconstructors) from serial dependence (29.1%–35.3% vs. 4.7%–19.4%) and tail-fit error (3.0%–8.1%), alongside quantile, block-length, and excess-kurtosis sensitivity audits.

II. Related Work

Dynamical Systems, Reconstruction, and Forecasting Detectors: Multi-sensor telemetry models system dynamics as state trajectories $\dot{x}(t) = \mathbf{f}(x(t))$ [10]. Neural ODEs [11] parameterize velocity fields in raw observation space, where integrating through high-frequency sensor noise induces numerical stiffness. Koopman theory and dynamic mode decomposition [12]–[14] learn linear operators on observable embeddings, identifying disruptions via spectral decoherence; our proposed Operator-Entropy JEPA architecture transfers this principle to a latent transition operator regularized by spectral entropy.

Reconstructive detectors model nominal patterns via observation reconstruction: LSTM forecasting with dynamic thresholding for spacecraft telemetry [3], association-discrepancy attention in Anomaly Transformer [4], 2D FFT-periodicity variation in TimesNet [5], and adversarial sub-series attention in TranAD [6]. These share a vulnerability: point-level losses bias models toward stochastic noise, accumulating residual variance across irrelevant channels in high-dimensional arrays.

Threshold Calibration, Contrastive, and Self-Supervised Detection: Unsupervised monitoring requires automated rules to map residual scores to binary alarms without labels. Non-parametric thresholding [3] and Streaming Peaks-Over-Threshold (SPOT) [7] fit Generalized Pareto tails under design risk q; however, EVT asymptotic guarantees assume i.i.d. samples, which temporal autocorrelation and operational drift violate.

To avoid reconstructing noise, contrastive methods train discriminative representations: NCAD [8] injects synthetic anomalies, yielding lower calibrated point detection than predictive models (0.058 vs. 0.160 macro Point-F1 across 11 datasets, Table II(d)). DCdetector [15] maximizes dual-attention consistency (0.070 macro Point-F1). Joint-Embedding Predictive Architectures [9], [16], [17] predict target representations in latent space with non-contrastive collapse prevention [18]–[20]. Time-series extensions [21] evaluate ranking or post-hoc threshold sweeps without studying how latent residuals behave under automated tail calibration.

Physics-Inspired Priors and Conceptual Positioning: PINNs [22] and Hamiltonian/Lagrangian networks [23], [24] embed physical conservation laws as rigid differential constraints; their stiff competing gradients [25] and known-PDE requirements fit poorly to uncalibrated multi-sensor telemetry. We strictly distinguish *physics-inspired* algebraic regularizers (where latent coordinates are abstract learned features, not physical state variables) from *physics-informed* conservation constraints; *causal data access* from *score-emission latency*; and *operator non-degeneracy* from *encoder representation diversity* and *anomaly separability*.

III. Multi-Sensor IoT and Remote Sensing Benchmark Suite

Our benchmark suite comprises 11 datasets spanning 45 continuous telemetry streams sourced from NASA spacecraft archives [3], the UCI Machine Learning Repository [26], [27], TimeEval [28], and mTSBench [29]: 33 streams are evaluated under the primary multi-seed held-out calibration protocol and 12 reserved for operational case studies. Streams span four physical domains: *spacecraft telemetry* (SMAP and MSL entities, with two short-nominal MSL streams reserved; SWAN-SF solar magnetograms [30]); *wearable kinematics* (six Daphnet FoG streams [26]: 64 Hz tri-axial accelerometry for Freezing-of-Gait events, so S=64 is a 1-s horizon); *industrial processes* (14 GHL gasoil heating-loop streams, the Genesis pick-and-place testbed, and GECCO water-distribution chemistry/hydrology at 1-minute resolution); and *smart infrastructure* (Room-Occupancy environmental sensing [31], six CICIDS edge network-flow streams, CalIt2 people-flow [27], and Metro I-94 highway traffic).

Each stream provides nominal training sequences and labeled anomaly test sequences; channels are standardized by training statistics with $\boldsymbol{\sigma}$ floored at 10^{-2} against flatline phases.

We evaluate the standardized subsets studied in the literature [3], [28] (of 55 SMAP and 27 MSL archive entities) for strict matching and reproducible runtime.

To eliminate circular in-sample threshold tuning, each stream's nominal sequence is partitioned chronologically into training (65%), nominal calibration (17.5%), and held-out nominal evaluation (17.5%) blocks separated by an unobserved 64-step isolation gap (Section V-A). Stream inclusion in the primary multi-seed held-out protocol was governed by two pre-specified criteria prior to model evaluation: (i) a nominal sequence long enough to yield at least $N_{\text{cal}} \geq 500$ calibration windows after the $65\%/17.5\%/17.5\%$ split, and (ii) tractable stride-1 multi-seed training and evaluation runtime. Exactly 33 streams across seven datasets satisfy both criteria (6 Daphnet, K=9; 14 GHL, K=16; 4 MSL, K=55; 6 SMAP, K=25; Genesis, K=18; Metro, K=5; SWAN, K=38). The remaining 12 streams (CalIt2, K=2; two Room-Occupancy, K=5; two short-nominal MSL; GECCO, K=9; and six 72-channel CICIDS network flows) are evaluated in operational incident case studies and in a broader 11-dataset, 130-stream single-seed baseline cross-validation panel (Table II(d)). The dyadic front-end ablation (Table I(b)) uses a five-stream panel that includes one diagnostic SMD stream (`machine-1-2`, K=38).

Two external-validity caveats bound interpretation. First, SMAP and MSL contain documented mislabeled segments and near-trivial point anomalies [32]; their per-dataset scores represent upper bounds on operational detectability. Second, because multiple streams within SMAP, MSL, Daphnet, and GHL share sensor hardware and environmental telemetry characteristics, streams within a dataset are not independent application domains. We therefore report unweighted Dataset-Macro means, per-dataset between-stream and within-stream seed standard deviations, leave-one-dataset-out (LODO) domain sensitivity, and dataset-clustered permutation tests (Table II(c)), and restrict conclusions strictly to the evaluated entities and sampling regimes rather than extrapolating to entire mission archives.

IV. Methodology: Operator-Entropy Joint-Embedding Predictive Architecture

A. Problem Formulation and Latent Space Prediction

Let $\mathbf{X} \in \mathbb{R}^{T\times K}$ denote a multivariate stream partitioned into context $x_{\text{ctx}} = \mathbf{X}[t-C:t,:] \in \mathbb{R}^{C\times K}$ and suspect horizon $x_{\text{tgt}} = \mathbf{X}[t:t+S,:] \in \mathbb{R}^{S\times K}$. Reconstructive baselines (TimesNet, TranAD, Anomaly Transformer) minimize pointwise error $\min_\theta \mathbb{E}\|\mathbf{X} - \mathcal{D}_\theta(\mathcal{E}_\theta(\mathbf{X}))\|_F^2$, which can weight channels with different noise scales unequally; whether this consumes model capacity is not tested here. We instead adopt the JEPA framework [9]: a context encoder E_θ maps $x_{\text{ctx}} \to z_{\text{ctx}} \in \mathbb{R}^D$ and an EMA momentum target encoder E_ϕ maps $x_{\text{tgt}} \to z_{\text{tgt}}$, with $\phi_k \leftarrow m_k\phi_{k-1} + (1-m_k)\theta_k$, $m_k \in [0.996, 0.9995]$. The objective forecasts target representations in latent space to filter out stochastic sensor noise and condition prediction residuals for automated tail calibration.

B. Latent Predictor Architectures and Dynamical Inductive Priors

We formulate Operator-Entropy JEPA as our primary architecture, and evaluate it alongside an unconstrained baseline (`TS-JEPA`) and two comparative physical formulations (Reynolds-Stress and Potential-Flow):

1) Operator-Entropy JEPA (Spectral Dynamics & Modal Dispersion): Motivated by Koopman operator theory [33], [34], where nonlinear trajectories are mapped via linear operators on observable functions, we parameterize the predictive latent transition via a learned operator $\mathbf{K} \in \mathbb{R}^{D\times D}$: $\hat{z}_{\text{tgt}} = \mathbf{K}z_{\text{ctx}}$. To prevent collapse onto a low-rank subspace (dimensional collapse) or to zero (energy collapse), the regularizer decouples spectral uniformity from total operator energy.

Let $\sigma_1 \geq \cdots \geq \sigma_D \geq 0$ denote singular values of $\mathbf{K}$, $\lambda_i = \sigma_i^2$ eigenvalues of $\mathbf{K}\mathbf{K}^\top$, and $\mathcal{E}(\mathbf{K}) = \|\mathbf{K}\|_F^2 = \sum_i \lambda_i$ total energy. To ensure collapsing modes ($\lambda_i \to 0^+$) receive strictly negative gradients ($\partial\Delta\mathcal{H}/\partial\lambda_i < 0$) without sign reversal, we form the smoothed modal distribution $p_i = (\lambda_i+\epsilon_{\log})/\sum_{j=1}^D(\lambda_j+\epsilon_{\log})$ ($\epsilon_{\log} = 10^{-12}$) and Von Neumann spectral entropy $\mathcal{H}_{\text{spectral}}(\mathbf{K}) = -\sum_{i=1}^D p_i \log p_i$. The *Spectral Entropy Deficiency* measures Kullback-Leibler divergence from the uniform equipartition $\mathbf{u} = [1/D,\ldots,1/D]^\top$:

$$\Delta\mathcal{H}(\mathbf{K}) = \log D - \mathcal{H}_{\text{spectral}}(\mathbf{K}) = \mathcal{D}_{\text{KL}}(\mathbf{p}\,\|\,\mathbf{u}) \geq 0, \quad (1)$$

where $\Delta\mathcal{H} = 0$ if and only if all modes carry equal energy ($\sigma_1 = \cdots = \sigma_D$), while rank-1 collapse yields $\Delta\mathcal{H} \approx \log D$. Minimizing $\Delta\mathcal{H}$ drives $\mathbf{K}$ toward isometric modal equipartition rather than low-rank truncation, acting directly on learned embeddings.

To penalize energy collapse ($\|\mathbf{K}\|_F \to 0$) without explosive gain or zero-gradient clamping, we impose a smooth shifted Bregman energy barrier relative to nominal isometric scale $\mathcal{E}_0 = \gamma D$ ($\gamma = 1.0$):

$$\Phi_{\text{energy}}(\mathbf{K}) = r(\mathbf{K}) - 1 - \log\big(r(\mathbf{K})+\epsilon\big) + \log(1+\epsilon) \geq 0, \quad (2)$$

where $r(\mathbf{K}) = \|\mathbf{K}\|_F^2/(\gamma D)$ and $\epsilon = 10^{-6}$. This barrier is C^∞ smooth for $r \geq 0$, vanishes at $r = 1$, caps zero-energy penalty at $\approx 12.8\,\beta_{\text{norm}}$, and has radial derivative $\partial\Phi_{\text{energy}}/\partial r|_{r=0} = 1 - 1/\epsilon < 0$. With $\mathbf{K} \sim \mathbf{I} + \mathcal{N}(\mathbf{0}, 0.01^2\mathbf{I})$, $r \approx 1$ and initial barrier penalty vanishes. The composite regularizer is:

$$\mathcal{R}_{\text{spectral}}(\mathbf{K}) = \Delta\mathcal{H}(\mathbf{K}) + \beta_{\text{norm}}\Phi_{\text{energy}}(\mathbf{K}), \quad (3)$$

and the training objective minimizes $\mathcal{L}_{\text{Op-Entropy}} = \frac{1}{BD}\sum_b \|\mathbf{K}z_{\text{ctx}}^{(b)} - z_{\text{tgt}}^{(b)}\|_2^2 + \lambda_{\text{ent}}\mathcal{R}_{\text{spectral}}(\mathbf{K}) + \beta_{\text{Lyap}}\mathcal{L}_{\text{Lyap}}$ ($\beta_{\text{norm}} = 1.0, \lambda_{\text{ent}} = 0.5, \beta_{\text{Lyap}} = 0.02$). The optional discrete Lyapunov defect aligns trace-normalized batch covariances on the transition operator, $\mathcal{L}_{\text{Lyap}} = \frac{1}{D}\|\mathbf{\Sigma}_{\text{pred}}/s_{\text{ctx}} - \mathbf{\Sigma}_{\text{tgt}}/s_{\text{tgt}}\|_F^2$ ($s_m = \operatorname{tr}\mathbf{\Sigma}_m/D + 10^{-6}$, $m \in \{\text{ctx},\text{tgt}\}$), where $\mathbf{\Sigma}_{\text{pred}}, \mathbf{\Sigma}_{\text{tgt}}$ are the empirical covariances of $\mathbf{K}\,\text{sg}(z_{\text{ctx}})$ and z_{tgt} over the batch. Shared representation stages are fixed across controls. $\mathcal{R}_{\text{spectral}}(\mathbf{K})$ and $\mathcal{L}_{\text{Lyap}}$ act on $\mathbf{K}$ only; encoder collapse is mitigated empirically by EMA target dynamics and stop-gradient operations [16], [35].

Proposition 1 (Operator Non-Triviality Under Finite-ϵ and Singular Barriers)**.** *Let* $\mathcal{L}_\epsilon(\mathbf{K}) = \frac{1}{BD}\sum_{b=1}^B \|\mathbf{K}z_{ctx}^{(b)} - z_{tgt}^{(b)}\|_2^2 + \lambda_{ent}\mathcal{R}_{spectral}(\mathbf{K})$ *with* $\lambda_{ent}, \beta_{norm} > 0$*, target batch energy* $E_{tgt} > 0$*, and empirical cross-covariance* $\mathbf{M} =$

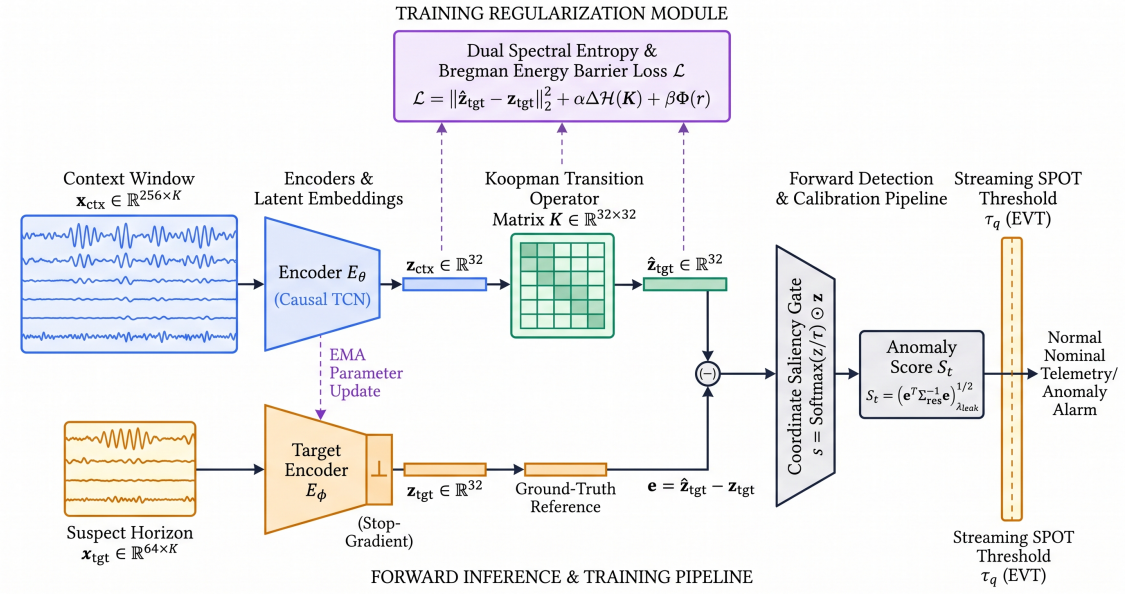


Fig. 1. Operator-Entropy JEPA architecture. Online (E_θ) and EMA target (E_ϕ) encoders map context ($\mathbf{x}_{\text{ctx}} \in \mathbb{R}^{256\times K}$) and suspect ($\mathbf{x}_{\text{tgt}} \in \mathbb{R}^{64\times K}$) windows into latent space, where a Koopman operator ($\mathbf{K} \in \mathbb{R}^{32\times32}$) forecasts $\hat{\mathbf{z}}_{\text{tgt}} = \mathbf{K}\mathbf{z}_{\text{ctx}}$ under spectral-entropy and Bregman-barrier regularization for coordinate-gated SPOT scoring.

$\frac{1}{BD}\sum_{b=1}^{B} z_{tgt}^{(b)} z_{ctx}^{(b)\top}$*: (i)* Finite-$\epsilon$ smooth regime ($\epsilon = 10^{-6}$): $\Phi_{energy}(\mathbf{0}) \approx 12.8$ *is finite and* $\partial\Phi_{energy}/\partial r|_{r=0} < 0$*, but the matrix gradient* $\nabla_{\mathbf{K}}\mathcal{R}_{spectral}(\mathbf{0}) = \mathbf{0}$ *vanishes because* $r(\mathbf{K}) \propto \|\mathbf{K}\|_F^2$*. The barrier alone produces no repulsive gradient at* $\mathbf{K} = \mathbf{0}$*; instead, whenever* $\mathbf{M} \neq \mathbf{0}$*,* $\nabla_{\mathbf{K}}\mathcal{L}_\epsilon(\mathbf{0}) = -2\mathbf{M} \neq \mathbf{0}$ *excludes* $\mathbf{K} = \mathbf{0}$ *as a stationary point or minimizer. (ii)* Singular-barrier limit ($\epsilon \to 0^+$): *For unshifted* $\mathcal{L}_0(\mathbf{K})$*,* $\lim_{\|\mathbf{K}\|_F\to0} \Phi_{energy}(\mathbf{K}) = +\infty$ *and* $\lim_{\|\mathbf{K}\|_F\to0} \mathcal{L}_0(\mathbf{K}) = +\infty$ *unconditionally, independent of* $\mathbf{M}$*.*

Proof. (i) For $\epsilon > 0$, Φ_{energy} and $\Delta\mathcal{H}$ are smooth. By the chain rule, $\nabla_{\mathbf{K}}\Phi_{\text{energy}}(\mathbf{K}) = \frac{2}{\gamma D}\Phi'_{\text{energy}}(r)\,\mathbf{K} = \mathbf{0}$ at $\mathbf{K} = \mathbf{0}$. At $\mathbf{K} = \mathbf{0}$, $\boldsymbol{\rho}(\mathbf{0}) = \mathbf{I}/D$ minimizes the nonnegative smooth entropy deficiency $\Delta\mathcal{H}$, so $\nabla_{\mathbf{K}}\Delta\mathcal{H}(\mathbf{0}) = \mathbf{0}$. Differentiating the prediction loss gives $\nabla_{\mathbf{K}}\mathcal{L}_\epsilon(\mathbf{0}) = -2\mathbf{M}$; if $\mathbf{M} \neq \mathbf{0}$, $\mathbf{K} = \mathbf{0}$ is non-stationary. (ii) For $\epsilon = 0$, $\Phi_{\text{energy}} = r - 1 - \log r \to +\infty$ as $r \to 0^+$; nonnegativity of the other terms yields $\lim_{\|\mathbf{K}\|_F\to0} \mathcal{L}_0(\mathbf{K}) = +\infty$. □

Remark (Scope, Cross-Covariance Verification, and Non-Collapse): Proposition 1 guarantees operator non-degeneracy, not representation diversity or anomaly detection. Unlike representation regularizers (VICReg [20], Barlow Twins [18], whitening [19]) that penalize covariance collapse of encoder outputs $z_{\text{ctx}}, z_{\text{tgt}}$, $\mathcal{R}_{\text{spectral}}(\mathbf{K})$ constrains only the transition spectrum. If E_θ maps all inputs to a constant, $\mathbf{K} = \mathbf{I}$ attains zero loss; conversely, an isometric operator ($\kappa(\mathbf{K}) \approx 1$) is compatible with representations in a low-dimensional subspace ($\text{RankMe}(z_{\text{ctx}})\approx5.3$, $\text{RankMe}(z_{\text{tgt}})\approx11.6$ of 32; Table I(c)). Empirically, target variance ($E_{\text{tgt}} \in [0.09, 5.88]$) and coupling ($\|\mathbf{M}\|_F > 0$) hold across all 195 primary runs because adjacent windows share LayerNorm-standardized features under cosine EMA $m_k \in [0.996, 0.9995]$ [35].

2) Auxiliary Quadratic Second-Moment Head: Inspired by Reynolds decomposition in fluid turbulence, which separates mean flow from velocity fluctuations, we evaluate an auxiliary quadratic second-moment head in latent space. The observed matrix $\boldsymbol{\Sigma}_{\text{obs}} = (\mathbf{P}z_{\text{tgt}})(\mathbf{P}z_{\text{tgt}})^\top + 10^{-3}\mathbf{I} \in \mathbb{S}_{++}^{d_s\times d_s}$ ($d_s = 16$, via projection $\mathbf{P} \in \mathbb{R}^{d_s\times D}$) is an instantaneous rank-1 projected outer product plus ridge penalty, not a physical turbulent covariance. A closure head outputs lower-triangular $\mathbf{L} \in \mathbb{R}^{d_s\times d_s}$ reconstructing predicted covariance $\boldsymbol{\Sigma}_{\text{pred}} = \mathbf{L}\mathbf{L}^\top + 10^{-3}\mathbf{I}$. The objective minimizes $\mathcal{L}_{\text{Reynolds}} = \text{MSE}(\hat{z}_{\text{tgt}}, z_{\text{tgt}}) + 0.1\ \text{MSE}(\boldsymbol{\Sigma}_{\text{pred}}, \boldsymbol{\Sigma}_{\text{obs}})$. At test time, scoring adds the second-moment discrepancy: $\mathcal{S}(t) = g(\hat{z}_{\text{tgt}} - z_{\text{tgt}}) + 0.5\,g(\text{vec}(\boldsymbol{\Sigma}_{\text{obs}} - \boldsymbol{\Sigma}_{\text{pred}}))$.

3) Potential-Flow JEPA (Conservative Field Theory and Flow Matching): Under Helmholtz-Hodge decomposition a smooth velocity field splits into a curl-free gradient and a solenoidal part. We parameterize $\hat{v} = -\nabla_{z_t}\Phi_\psi$ via a residual MLP conditioned on state z_t, flow time $t \in [0, 1]$, context z_{ctx}, and a regime projection selected among $K_r = 4$ orthonormal subspaces (d=8, D=32) with tangent projection $v_\perp = v - \langle u, v\rangle u$, $u = z_t/\|z_t\|_2$. Training uses conditional flow matching along linear paths from noise $z_{\text{noise}} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ to z_{tgt}, $\mathcal{L}_{\text{Pot-Flow}} = \mathbb{E}_{t,z_{\text{noise}}}\|\hat{v} - (z_{\text{tgt}} - z_{\text{noise}})\|_2^2$, and inference queries the field at the trajectory midpoint, $\mathbf{e}(t) = \hat{v}(0.5z_{\text{tgt}}, 0.5, z_{\text{ctx}}, p_{\text{regime}}) - z_{\text{tgt}}$. The curl-free bias is hypothesized to penalize cyclic nominal orbits and degrade detection on periodic kinematics (Section VII-B).

C. Anomaly Scoring and Operational Detection Latency

During inference, windows $x_{\text{ctx}} = \mathbf{X}[t - C : t, :]$ and $x_{\text{tgt}} = \mathbf{X}[t : t + S, :]$ are processed at stride 1. For TS-JEPA and Potential-Flow, the score gates the residual $\mathbf{e}(t) = z_{\text{tgt}}(t) - \hat{z}_{\text{tgt}}(t)$ (or whitened $\boldsymbol{\Omega}^{1/2}(\mathbf{e}(t) - \boldsymbol{\mu}_r)$) via $g(\cdot)$ (Section IV-D1): $\mathcal{S}(t) = g(\mathbf{e}(t))$. Because final LayerNorm projects embeddings to radius $\sqrt{D}$, Operator-Entropy JEPA tracks pre-LayerNorm radial energy scaled by window RMS

$(e_{\text{ctx}}, e_{\text{tgt}})$ with $e_{\text{pred}} = e_{\text{ctx}}\|\hat{z}_{\text{tgt}}\|_2/(\|z_{\text{ctx}}\|_2 + 10^{-6})$, scoring $\mathcal{S}(t) = g(\mathbf{e}(t)) + 0.2\,|e_{\text{tgt}}(t) - e_{\text{pred}}(t)|/(e_{\text{tgt}}(t) + e_{\text{pred}}(t) + 10^{-6})$. For Reynolds-Stress, $\mathcal{S}(t) = g(\mathbf{e}(t)) + 0.5\,g(\text{vec}(\mathbf{\Sigma}_{\text{obs}} - \mathbf{\Sigma}_{\text{pred}}))$. Shared stages are fixed across controls (Section VI-A).

Three-Part Operational Causality Framework: To resolve the methodological confounding between data availability, decision latency, and evaluation attribution, we formalize detection under three operational criteria:

(a) **Data causality ($\mathcal{F}_\tau$-measurability):** At decision time $\tau = t + S - 1$, the model observes only history $\mathbf{X}_{:\tau} = \mathbf{X}[t - C : \tau, :]$. Neither encoder accesses future observations beyond τ.

(b) **Score-emission latency ($S-1 = 63$ steps):** Because suspect representation z_{tgt} encodes the horizon $\mathbf{X}[t : t+S, :]$, window score s_t cannot be computed until timestep $\tau = t+S-1$. This imposes an inherent emission latency of $S-1 = 63$ steps: 0.98 s on Daphnet (64 Hz), 10.5 min on SMAP, MSL, and GHL under standard 10-s (0.1 Hz) sampling [3], [29], 63 min on GECCO (1 sample/min), and 63 h on Metro (1 sample/h). This latency guarantees zero future lookahead, but must not be confused with instantaneous detection at fault inception: batch scoring over horizon S=64 fits supervisory health monitoring and slow subsystem drift where anomaly persistence distinguishes faults from noise, but cannot provide sub-second emergency tripwires on low-frequency telemetry. Section VII-D quantifies how shortening S trades this latency against false-alarm control.

(c) **Alarm attribution convention:** We distinguish two attribution regimes: (1) *strictly causal trailing attribution*, where score s_t is emitted and evaluated strictly at τ, providing zero-lookahead online monitoring; and (2) *retrospective buffered smear attribution*, where s_t is attributed across the entire suspect window $[t+1, t+S]$. Under unadjusted point evaluation, buffered alarms in $[t^* - S, t^*]$ register as false alarms: buffering provides fixed-lag detection, not early warning. The two regimes represent distinct operational realities and are reported separately throughout (Table II(a),(b)).

D. Shared Internal Inductive Biases

To stabilize learning across high-dimensional, noisy telemetry, the latent JEPA architectures incorporate shared inductive mechanisms:

1) Coordinate Saliency Gating and the Auxiliary Channel-Gate Variant: On high-dimensional streams ($K \geq 25$), faults frequently originate in a localized subsystem while the remaining channels exhibit ambient noise, diluting localized anomalies in isotropic representations. The deployed detector mitigates this in latent space via an anti-drowning coordinate saliency gate $g(\mathbf{e}) \equiv \mathcal{D}_{\text{sal}}(\mathbf{e})$ on error vectors $\mathbf{e} = \hat{z}_{\text{tgt}} - z_{\text{tgt}} \in \mathbb{R}^D$:

$$\mathcal{D}_{\text{sal}}(\mathbf{e}) = (1-\alpha)\|\mathbf{e}\|_2 + \alpha\sqrt{D\sum_{d=1}^{D} w_d e_d^2}, \qquad (4)$$

$$w_d = \frac{\exp(|e_d|/\tau)}{\sum_{j=1}^{D}\exp(|e_j|/\tau)},$$

amplifying sparse coordinate deviations by up to $\sqrt{D}$. We fix blend parameter $\alpha = 0.5$ and softmax temperature $\tau = 0.5$ across all latent predictors. Reconstructive baselines (TimesNet, TranAD) evaluate unweighted observation errors and do not use the gate. This is the discrepancy gate used in every latent experiment: input channels are encoded unweighted, and the gate acts only on the D-dimensional predictive residual.

a) Auxiliary Input-Channel Gate (Evaluated Variant): A strictly causal gate weights each channel by the sigmoid of its standardized context-window variance share, $g_k = \sigma((s_k - \bar{s})/(\sigma_s + \epsilon))$, applied symmetrically to both encoders so that $\partial g_k/\partial \mathbf{x}_{\text{tgt}} \equiv \mathbf{0}$ (identity at K=1). It cannot isolate quiet faults (recall 0.02–0.07 under every gating regime, Section VI); the deployed pipeline weights inputs uniformly.

2) Multiscale Frequency Front-End: Anomalies span temporal scales from sensor spikes to thermal drifts; the encoder front-end therefore splits each channel into J=4 smooth dyadic shells on an $N_{\text{fft}} = 256$ rFFT grid, normalized bin-wise into an exact amplitude partition of unity, $\sum_j \Psi_j[\omega_m] \equiv 1$ (measured deviation $< 1.2 \times 10^{-7}$). Each shell feeds a causal 1D convolution ($k \in \{15, 9, 5, 3\}$ coarse-to-fine), recombined by learned scale routing. Because the shells overlap smoothly at band edges, the front-end is *less* frequency-localized than standard wavelet banks. We therefore describe it as a Fourier-domain multiscale front-end, not an exact Littlewood–Paley octave decomposition, and measure its empirical contribution by matched ablation (Table I(b)).

3) Kurtosis-Adaptive Extreme Value Theory (EVT) and Calibration Realities: Binary alarms $\hat{y}_t = \mathbb{I}(\mathcal{S}(t) > \tau_t)$ are generated via Streaming Peaks-Over-Threshold (SPOT) [7]: excesses over a high threshold u (98th percentile) asymptotically follow a Generalized Pareto Distribution (GPD) with scale σ and shape ξ:

$$\tau = u + \frac{\sigma}{\xi}\left[\left(\frac{qN}{N_u}\right)^{-\xi} - 1\right], \qquad (5)$$

where $q = 10^{-3}$ denotes the design risk probability, N is total calibration samples, and N_u is the count of excesses. To test whether sample excess kurtosis γ_2 of calibration residuals can compensate for pre-asymptotic tail heaviness, we evaluate two heuristic adjustments: (i) scaling the total threshold from zero, $\tau_{\text{eff}} = \tau[1 + \frac{1}{6}\tanh(\gamma_2)]$, and (ii) scaling strictly the tail excess above u, $\tau_{\text{exc}} = u + (\tau - u)[1 + \frac{1}{6}\tanh(\gamma_2)] \geq u$.

These multipliers are engineering heuristics for pre-asymptotic tails, not EVT theorems. The primary pipeline uses unadjusted SPOT ($u = p_{98}, q = 10^{-3}$) and evaluates both multipliers as sensitivity arms (Section VII-F). Because 64-step buffered smoothing produces tightly concentrated, platykurtic scores ($\gamma_2 < 0$) on 31.9% of TS-JEPA cells, scaling total τ pulls τ_{eff} below u into the bulk distribution (6.40% $\rightarrow$ 16.30% buffered FPR), whereas scaling only the excess $(\tau - u)$ preserves the $u = p_{98}$ floor (6.33% buffered FPR).

V. Experimental Protocol and Evaluation Metrics

A. Experimental Setup and Hyperparameters

To ensure rigorous and fair comparison, all architectures are trained and evaluated under an identical streaming time-series pipeline. Telemetry streams are processed using a sliding window with past context duration $C = 256$ and suspect target horizon $S = 64$ (total window length $L = 320$), evaluated with a stride of 1 timestep across all testing streams.

Context representations $z_{\text{ctx}} \in \mathbb{R}^D$ ($D = 32$) are extracted via a multiscale dyadic front-end and causal dilated TCN (6 residual blocks, F=48, kernel 5, dilations $\{1, 2, 4, 8, 16, 32\}$); multi-statistic pooling (terminal, mean, max, and channel statistics) projects the C=256 context through a linear bottleneck recording pre-LayerNorm radial energy before LayerNorm and dropout (p=0.20). The momentum target encoder uses cosine EMA m_{base}=0.996 $\rightarrow$ m_{max}=0.9995.

Models are optimized using AdamW with an initial learning rate $\eta = 10^{-3}$ decayed to $\eta_{\text{min}} = 10^{-5}$ via Cosine Annealing, weight decay 10^{-4}, and mini-batch size $B = 32$ across 20 training epochs on nominal telemetry sequences.

a) Held-Out Calibration Protocol: To evaluate out-of-sample threshold calibration without circularity, each stream's nominal telemetry is partitioned into three disjoint chronological blocks: *training* (first 65%), *nominal calibration* (next 17.5%), and *nominal evaluation* (final 17.5%, separated by an unobserved 64-step isolation gap). The training block optimizes representation weights alone. Withheld from training, the nominal calibration block fits per-stream GPD tail parameters σ, ξ above u=0.98 at q=10^{-3}, estimating γ_2 strictly within this interval. The nominal evaluation block then measures held-out FPR on telemetry unobserved during threshold selection. Thresholds remain fixed during test evaluation to measure static EVT stability under drift; dynamic rolling SPOT is evaluated separately in the operational incident analysis (Section VII-C).

b) Calibrator Assignment: All experiments use standard SPOT ($u = p_{98}$, $q = 10^{-3}$, no kurtosis multiplier) on held-out nominal splits, except the rolling-SPOT incident analysis and the calibrator comparison itself (Table IV); no table mixes calibrators across models.

c) Multi-Seed Replication, Primary Endpoints, and Equivalence Testing: Principal models on the 19 non-GHL streams run five seeds (42, 123, 456, 7, 2024); the 14 GHL streams ($> 10^6$ steps) and ablation controls run three seeds (five on the two smallest GHL streams). We report held-out nominal FPR, Point-F1, and PR-AUC as distinct outcomes; nonsignificant differences are not evidence of parity. Inference reports stream-macro means, seed SDs, moving-block bootstrap (L=64) per-cell FPR intervals, stream-resampled paired CIs (2,000 iterations, seeds averaged within stream, n=19 or n=33), dataset-clustered checks (N=7), and Benjamini–Hochberg FDR-adjusted Wilcoxon signed-rank tests (q_{FDR}=0.05). For physical regularizer ablations, we pair two-sided tests with Two One-Sided Tests (TOST): H_{01} : $\mu_1 - \mu_2 \leq -\Delta$ (p_{lower}) and H_{02} : $\mu_1 - \mu_2 \geq +\Delta$ (p_{upper}; $p_{\text{TOST}} = \max(p_{\text{lower}}, p_{\text{upper}})$) alongside 90% ($1-2\alpha$ at α=0.05) and 95% CIs. We prespecify $\Delta = \pm 0.05$ Point-F1 for cross-architecture contrasts ($\approx$ 25% of baseline F1, $< 0.15\times$ between-stream SD) and $\Delta = \pm 0.03$ for fixed-backbone regularizer controls; equivalence requires $p_{\text{TOST}} < 0.05$ and the 90% CI strictly inside $[-\Delta, +\Delta]$.

B. Baseline Architectures, Fairness Checks, and Cohort Structure

All models share a common streaming pipeline (C=256, S=64, stride 1, AdamW η=$10^{-3} \rightarrow 10^{-5}$, weight decay 10^{-4}, batch 32, 20 epochs, SPOT u=p_{98}, q=10^{-3}). Three fairness checks bound this protocol: (i) *Training-budget sensitivity* (7 streams $\times$ 3 seeds): doubling optimization from 10 to 20 epochs raises JEPA mean F1 from 0.038 to 0.048 (OpEntropy 0.029 $\rightarrow$ 0.065) while reconstructor F1 is stable (0.158 $\rightarrow$ 0.154; TimesNet 0.154 $\rightarrow$ 0.151, TranAD 0.161 $\rightarrow$ 0.158) and held-out FPR stays 0.087–0.092 vs. 0.333. (ii) *Native published windows* (5 streams): at native windows (TranAD W=10, TimesNet W=96), F1 rises (0.164, 0.218), but observation residuals remain leptokurtic (γ_2 = 448.25 and 98.14 vs. 3.14 for TS-JEPA) with high nominal FPR (28.4%, 44.1%). (iii) *Published-score and 130-stream cross-check*: baselines replicate published PA-F1 on SMD/MSL, and Table II(d) evaluates 130 streams across 11 datasets against NCAD (149,634 parameters at K=25), DCdetector (298,400), and Anomaly Transformer (4,785,120). **Cohort Summary and Reproducibility Specification:** Five cohorts address distinct questions: (1) *33-stream primary suite* (19 non-GHL + 14 GHL, Tables II(a)–(c), IV(a)); (2) *19-stream non-GHL mechanism/operating suite* (12 arms, Table I(a),(e), Table III(b), Fig. 2); (3) *6-stream factorial and fault suite* (Table I(c),(d)); (4) *5-stream horizon and front-end panels* (Table I(b), Table III(a)); and (5) *130-stream 11-dataset panel* (Table II(d)). Parameter counts at K=25: OpEntropy 285,701, TS-JEPA 289,061, Reynolds 296,037, Potential-Flow 338,209, TimesNet 648,553, TranAD 205,746. Non-finite values are zero-imputed ($\boldsymbol{\sigma} \geq 10^{-2}$); SPOT fits GPD (ξ, σ) above u=p_{98} via Grimshaw MLE with L-BFGS-B fallback ($\xi \in [-0.9, 10], \sigma \geq 10^{-6}$). All 195 primary runs converged on PyTorch 2.4 / CUDA 12.4 (RTX 5090, 0.60–0.84 ms/window, 183.8–185.3 MB VRAM); per-stream manifests, thresholds, and code are at https://github.com/quaid281/NCAD.

C. Multi-Tier Statistical Evaluation Metrics

Critical surveys [28], [36] document severe reporting vulnerabilities in this literature (point-adjustment inflation, channel-averaging imbalances, threshold gaming), so we report metrics across complementary tiers. *Exploratory operating characteristics*: event recall over nominal calibration budgets and over a realized-test-FPR interval $[0.10, 0.50]$, interpolated within stream–seed curves only where both bounds are observed. *Secondary endpoints*: Dataset-Macro Point-F1 (unweighted mean across datasets, preventing the 14-stream GHL block from dominating); channel-macro Point-F1; PR-AUC (average precision, ranking quality independent of calibration); event recall (fraction of contiguous anomaly segments containing $\geq$ 1 alarm); mean detection delay Δt (onset-to-first-alarm

lag over detected events only); and the trivial all-positive reference $\mathrm{F1}_{\text{all-pos}} = 2\pi/(1+\pi)$ at base rate π; models below it underperform a constant alarm. Pooled micro-F1 exposes flood-prone models; contiguous PA-F1 and test-sweep PA-F1 quantify the residual calibration gap.

VI. Ablation Studies and Analysis

A. Ablation 1: Physical Regularizer Controls on Fixed Architectures

To test whether the physics-inspired regularizers benefit detection, Table I(a) reports controlled ablations across all 19 non-GHL streams (principal models at five seeds, controls at three, under the held-out protocol), disabling each physical term on a fixed backbone. Because controls run three seeds against the five-seed principal runs, the paired fixed-backbone means below are computed on the shared three-seed subset (n=19 streams) with 90% (TOST α=0.05) and 95% bootstrap CIs and individual one-sided Wilcoxon TOST p-values ($p_{\text{lower}}, p_{\text{upper}}$; $p_{\text{TOST}} = \max(p_{\text{lower}}, p_{\text{upper}})$):

1) **Operator-Entropy vs. λ_{ent}=0 (fixed Koopman backbone):** Disabling the spectral regularizer leaves Point-F1 unchanged (0.232 ± 0.039 vs. 0.235 ± 0.038; difference -0.004, 90% CI $[-0.011, +0.003]$, 95% CI $[-0.014, +0.004]$, two-sided Wilcoxon p=0.89; PR-AUC 0.365 vs. 0.365), confirming formal TOST equivalence within $\Delta = \pm 0.03$ ($p_{\text{lower}} = 0.00074, p_{\text{upper}} = 6.6 \times 10^{-5}, p_{\text{TOST}} = 0.00074$) and $\Delta = \pm 0.05$ ($p_{\text{TOST}} = 7.7 \times 10^{-5}$).
2) **Reynolds-Stress vs. α_{stress}=0 (fixed backbone):** Disabling the stress loss yields 0.195 ± 0.029 vs. 0.182 ± 0.036 ($+0.013$, 90% CI $[-0.001, +0.030]$, 95% CI $[-0.003, +0.034]$, two-sided p=0.59), confirming TOST equivalence within $\Delta = \pm 0.05$ ($p_{\text{lower}} = 6.6 \times 10^{-5}, p_{\text{upper}} = 0.0019, p_{\text{TOST}} = 0.0019$).
3) **Potential-Flow vs. no-regime (K_r=1):** Disabling regime routing yields 0.135 ± 0.062 vs. 0.189 ± 0.024 (-0.054, 90% CI $[-0.131, +0.012]$, 95% CI $[-0.148, +0.022]$, two-sided p=0.86), failing equivalence at $\Delta = \pm 0.05$ ($p_{\text{lower}} = 0.17, p_{\text{upper}} = 0.0090, p_{\text{TOST}} = 0.17$).

We strictly distinguish these fixed-backbone regularizer controls from cross-architecture comparisons against unconstrained TS-JEPA (which uses an MLP predictor without radial energy scoring or $\mathcal{L}_{\text{Lyap}}$): on the 19 non-GHL subset, OpEntropy JEPA vs. TS-JEPA yields $+0.028$ Point-F1 (90% CI $[-0.018, +0.074]$, 95% CI $[-0.026, +0.085]$, two-sided p=0.39), which is statistically inconclusive at $\Delta = \pm 0.05$ because the upper bound exceeds $+0.05$ ($p_{\text{lower}} = 0.0027, p_{\text{upper}} = 0.11, p_{\text{TOST}} = 0.11$; Reynolds vs. TS-JEPA -0.015, 90% CI $[-0.056, +0.026]$, $p_{\text{TOST}} = 0.11$), whereas across all 33 primary streams OpEntropy JEPA vs. TS-JEPA ($+0.016$, 90% CI $[-0.009, +0.045]$) and Reynolds vs. TS-JEPA (-0.008, 90% CI $[-0.031, +0.016]$) satisfy TOST(± 0.05) ($p_{\text{TOST}} = 0.0019$ and 0.0017).

B. Ablation 2: Factorial Isolation of the Entropy Term and Energy Barrier

A single on/off control cannot separate the spectral-entropy term from the energy barrier or show how either scales. Table I(c) crosses the two terms (neither, entropy only, barrier only, both) and sweeps $\lambda_{\text{ent}} \in \{0.05, 0.10, 0.50, 1.00, 2.00\}$ with the barrier on, on six streams at three seeds with a D=32 operator. Each arm reports operator rank $\mathrm{RankMe}(\mathbf{K})$, condition number $\kappa(\mathbf{K})$, representation rank $\mathrm{RankMe}(z_{\text{ctx}})/\,\mathrm{RankMe}(z_{\text{tgt}})$, the coefficient of variation $\mathrm{CV}_{\mathcal{L}}$ of the training loss as a stability proxy, and held-out calibration and detection outcomes in both attribution regimes.

Three observations follow. First, the unconstrained operator is already near-isotropic ($\mathrm{RankMe}(\mathbf{K})$=31.53 of 32, κ=1.437) because $\mathbf{K}$ is initialized near the identity; the regularizers tighten conditioning monotonically with λ_{ent} (31.85 and κ=1.255 at the deployed λ_{ent}=0.50, 31.95 and κ=1.154 at λ_{ent}=2.00) and lower $\mathrm{CV}_{\mathcal{L}}$ from 0.565 to 0.545. Second, because $\mathcal{L}_{\text{Lyap}}$ detaches z_{ctx}, the encoder representation is unaffected by λ_{ent} and β_{norm}: $\mathrm{RankMe}(z_{\text{ctx}})$ stays within 5.16–5.54 and $\mathrm{RankMe}(z_{\text{tgt}})$ within 11.54–11.58 of 32 in every arm. Operator non-degeneracy, representation diversity, and anomaly separability are therefore separate properties, and only the first moves systematically. Third, no arm or strength produces a large detection shift: buffered Point-F1 spans 0.032–0.047 and causal Point-F1 0.004–0.011 with buffered FPR tightly grouped in 0.070–0.074 and causal FPR in 0.016–0.017, while the deployed setting (λ_{ent}=0.50, β_{norm}=1) attains the highest causal event recall (0.444 vs. 0.369 unconstrained). The entropy term is a well-posed constraint that preserves operator conditioning without inflating false alarms.

Decoupling. On an eight-stream exploratory panel (two seeds, strictly causal trailing), coupling the detached trace-normalized Lyapunov defect (β_{Lyap}=0.02) with the smooth Bregman barrier (α_Z=0) cuts mean detection delay from 80.2 steps (historical one-sided clamp, β_{Lyap}=0) and 127.0 steps (unconstrained TS-JEPA) to 58.2 steps, while raising causal Point-F1 from 0.044–0.046 to 0.094, PR-AUC from 0.153–0.157 to 0.190, and event recall from 0.206–0.250 to 0.258 at causal FPR 0.013 (buffered Point-F1 0.116 vs. 0.063 for TS-JEPA). Adding explicit representation-covariance entropy (α_Z=0.005, β_{Lyap}=0.02) yields intermediate Point-F1 (0.068) and delay (60.7 steps), confirming that detaching z_{ctx} in $\mathcal{L}_{\text{Lyap}}$ aligns the operator without distorting encoder representations.

C. Ablation 3: Auxiliary Input-Channel Gating and Quiet-Fault Reference Monitors

The auxiliary causal input-channel gate is compared against uniform pooling, a non-causal target-only leaky control, and independent context/target gating under an identical TS-JEPA configuration (seed 42, four streams). Macro Point-F1 ranks uniform weighting (0.118) above context gating (0.077), with the leaky control at 0.128 and independent gating weakest (0.071); the effect is stream-dependent in sign (MSL `M-1`: $0.047 \rightarrow 0.227$; SMAP `P-3`: $0.280 \rightarrow 0.019$). The gate is an

evaluated variant; the deployed pipeline retains uniform input weighting.

Table I(d) injects six single-channel fault classes into six multivariate streams. The variance gate roughly doubles point recall relative to uniform pooling (macro 0.045 vs. 0.023) at about $3.6\times$ the nominal FPR (0.069 vs. 0.019), and a median-absolute-deviation variant is marginally stronger (0.060 recall). Quiet faults (flatline, stuck value, dropout) stay weakly detected under every latent gating regime (point recall 0.02–0.07). Two observation-space reference monitors test whether a complementary score can cover this blind spot: a rolling channel-variance floor (`lowvar`) and a maximum channel z-deviation (`dev_maxz`). The variance floor reaches point recall 0.18 (0.32 on mid-variance flatlines) and the z-deviation monitor 0.09–0.11, but at nominal FPR of 0.14–0.16, roughly eight times that of the latent detector. A complementary variance monitor recovers part of the quiet-fault gap only at a substantial alarm cost, so quiet-fault coverage remains an open limitation.

D. Ablation 4: Multiscale Dyadic-Shell Front-End

Replacing the standard front-end with the Fourier-domain dyadic shells (including the corrected Nyquist plateau shell carrying 40.5% of broadband white-noise energy and 76.0% ideal-band mass) plus per-shell causal convolutions and learned routing yields a mixed, stream-dependent trade-off (Table I(b)): across a five-stream exploratory panel under kurtosis-adaptive SPOT, macro PR-AUC rises from 0.213 to 0.263 ($+0.050$, led by SMD $+0.193$ and GECCO $+0.121$) and validation residual kurtosis drops from 1.409 to 1.069, while macro Point-F1 shifts from 0.118 to 0.078 (-0.040) due to threshold tightening on SMAP `P-3` ($0.301 \rightarrow 0.115$ as FPR drops $0.043 \rightarrow 0.012$). The block adds $\approx 1.6 \times 10^5$ parameters and shell overlap prevents strict octave isolation, so it acts as a ranking-conditioning front-end rather than a uniform F1 gain.

E. Mechanism-Discrimination Study: Latent Space vs. Observation Space

To isolate the operational mechanisms governing tail conditioning and threshold transfer, all 19 non-GHL streams were evaluated under the primary held-out protocol (five seeds, identical windows, AdamW budget, and SPOT u=0.98/q=10^{-3} calibration) across 12 controlled backbone arms (Table I(e), Fig. 2(a); neural variants sharing the `HybridTCNEncoder` with C=256, S=64, d=32, alongside linear and non-parametric controls).

Decoupling Score Compression from Representation Learning: Contrasting these controlled backbones (Table I(e) and Fig. 2(a)) separates score-distribution compression from learned representation dynamics: (1) *Unnormalized observation forecasting (`TCN-ObsPred`):* In raw observation space, residual excursions exhibit severe causal tail kurtosis ($\gamma_2 \approx 6.10$), pronounced calibration-to-holdout Kolmogorov–Smirnov distribution drift ($D_{\text{KS}} = 0.463$), and severe causal nominal false alarms (0.247). (2) *Observation forecasting with LayerNorm (`TCN-ObsPred-LN`):* Introducing a D=32 LayerNorm bottleneck mechanically curtails residual excursions, compressing causal kurtosis to $\gamma_2 \approx 0.91$ and halving causal FPR to 0.117 (buffered 0.193). However, calibration-to-holdout drift persists at $D_{\text{KS}} = 0.376$. Bounded geometry alone halves false alarms through score compression, but cannot resolve distribution drift across temporal blocks. (3) *Latent projection and representation drift suppression:* Shifting prediction into learned latent coordinates fundamentally alters drift. Unconstrained latent prediction (`TS-JEPA`) reduces calibration-to-holdout KS drift ($0.376 \rightarrow 0.214$, paired Wilcoxon $p = 0.0010$, $\gamma_2 = 1.33$; $0.376 \rightarrow 0.209$, $p = 0.0004$ for `OpEntropy`) and suppresses causal false alarms an additional five- to eight-fold ($0.117 \rightarrow 0.024$, event recall 0.487). (4) *Foresight vs. geometric compression:* Autoencoding context representations without temporal foresight (`JEPA-CtxRecon`) matches this drift reduction ($D_{\text{KS}} = 0.203$, $\gamma_2 = 2.59$) and achieves causal FPR 0.017 with event recall 0.542. Future temporal extrapolation is therefore not uniquely necessary for score conditioning; latent coordinate compression drives tail stability. (5) *Linear vs. non-parametric controls:* A linear projection forecaster (`PCA-LinPred-LN`) achieves $\gamma_2 = 0.95$, $D_{\text{KS}} = 0.233$, causal FPR 0.021, and event recall 0.452, whereas non-parametric isolation trees (`IsoForest`) collapse in event recall (0.149 at SPOT; 0.317 at f^*=0.01). Representation rank diagnostics confirm z_{ctx} and z_{tgt} occupy 5.6–6.1 and 8.5–14.0 of 32 dimensions, demonstrating that while LayerNorm geometry mechanically halves false alarms by bounding kurtosis, learned latent projection is strictly required to halve distribution drift and suppress false alarms to the 0.9%–2.8% operational regime.

VII. Results and Discussion

A. Benchmark Performance on Continuous Telemetry

Table II reports the comparison across 33 continuous telemetry streams under the held-out calibration protocol of Section V-A (SPOT, $u = 0.98, q = 10^{-3}$), with buffered (a) and strictly causal (b) attribution reported separately. The principal finding is relative false-alarm suppression; the evidence for general detection superiority is inconclusive.

Relative False-Alarm Suppression on Held-Out Telemetry: Latent predictive scoring via Operator-Entropy JEPA and TS-JEPA improves relative false-alarm control under static calibration: on 19 non-GHL streams, latent predictors achieve held-out nominal FPR 0.074–0.101 (95% stream-bootstrap CIs: TS-JEPA 0.093 $[0.048, 0.145]$, OpEntropy 0.101 $[0.043, 0.180]$) versus 0.300 $[0.162, 0.446]$ for TimesNet and 0.307 $[0.177, 0.445]$ for TranAD (three- to four-fold reduction buffered, and 8–20$\times$ causal: 0.011–0.028 vs. 0.222–0.243). Across all 33 streams the paired buffered reduction is significant: OpEntropy JEPA achieves -0.144 vs. TimesNet (CI $[-0.241, -0.065]$, $p = 0.006$) and -0.142 vs. TranAD ($p = 0.006$); TS-JEPA yields -0.153 ($p = 0.002$) and -0.151 ($p = 0.003$). Dataset-clustering ($N = 7$) yields $p = 0.016$–0.031, and leave-one-dataset-out (LODO) folds hold macro buffered FPR to 0.038–0.070 (OpEntropy) and 0.038–0.061 (TS-JEPA) vs. 0.084–0.183 (reconstructors). On MSL, TimesNet records lower buffered FPR than OpEntropy

TABLE I
ABLATION AND MECHANISM-DISCRIMINATION STUDIES: (A) FIXED-BACKBONE REGULARIZER CONTROLS (n=19, 5 SEEDS), (B) DYADIC FRONT-END (n=5), (C) OPERATOR REGULARIZER SWEEP (n=6, 3 SEEDS), (D) INJECTED QUIET-FAULT POINT RECALL (n=6), AND (E) 12-ARM BACKBONE COMPARISON (n=19, 5 SEEDS).

(a) Regularizer Controls on Fixed Architectures

Model / Control	Point-F1 (buffered)	PR-AUC	Held-out FPR	Event Recall
TS-JEPA (Unconstrained)	0.206 ± 0.025	0.331	0.093	0.477
Operator-Entropy JEPA	0.233 ± 0.028	0.351	0.101	0.550
$\lambda_{\text{ent}} = 0$ control	0.235 ± 0.038	0.365	0.099	0.580
Reynolds-Stress JEPA	0.190 ± 0.022	0.313	0.074	0.457
$\alpha_{\text{stress}} = 0$ control	0.182 ± 0.036	0.299	0.067	0.472
Potential-Flow JEPA	0.138 ± 0.044	0.235	0.087	0.449
no-regime control ($K_r = 1$)	0.189 ± 0.024	0.279	0.084	0.431
TimesNet (Baseline)	0.197 ± 0.018	0.329	0.300	0.583
TranAD (Baseline)	0.130 ± 0.004	0.287	0.307	0.490

(b) Dyadic-Shell Front-End On/Off

Stream (F1 / PR-AUC)	Standard Conv	Dyadic Shells	ΔF1 / ΔPR
SMAP `P-3`	**0.301** / 0.280	0.115 / **0.296**	-0.186 / $+0.017$
MSL `M-1`	0.014 / **0.491**	**0.016** / 0.449	$+0.002$ / -0.043
SMD `machine-1-2`	0.000 / 0.028	**0.068** / **0.221**	$+0.068$ / $+0.193$
Daphnet `S01R01E1`	**0.179** / **0.103**	0.130 / 0.066	-0.048 / -0.037
GECCO `water_quality`	**0.097** / 0.161	0.062 / **0.282**	-0.035 / $+0.121$
Macro Mean	**0.118** / 0.213	0.078 / **0.263**	-0.040 / $+0.050$

(c) Factorial and Strength Sweep of the Operator Regularizer

			Operator / representation / stability				Buffered			Causal		
Arm	λ_{ent}	β_{norm}	RankMe($\mathbf{K}$)	$\kappa(\mathbf{K})$	RankMe($z_{\text{ctx}}/z_{\text{tgt}}$)	$\text{CV}_{\mathcal{L}}$	FPR	F1	ER	FPR	F1	ER
Neither	0	0	31.53	1.437	5.16/11.57	0.565	0.071	0.044	0.250	0.017	0.009	0.369
Entropy only	0.5	0	31.84	1.256	5.25/11.57	0.550	0.071	0.040	0.266	0.017	0.011	0.389
Barrier only	0	1	31.54	1.431	5.37/11.58	0.563	0.074	0.047	0.266	0.016	0.011	0.389
Both	0.05	1	31.61	1.395	5.16/11.57	0.562	0.074	0.039	0.250	0.017	0.008	0.373
Both	0.10	1	31.67	1.363	5.19/11.58	0.559	0.073	0.039	0.250	0.017	0.008	0.421
Both (deployed)	0.50	1	31.85	1.255	5.54/11.56	0.549	0.072	0.038	0.266	0.017	0.010	0.444
Both	1.00	1	31.91	1.204	5.25/11.55	0.546	0.070	0.037	0.250	0.017	0.007	0.365
Both	2.00	1	31.95	1.154	5.34/11.54	0.545	0.072	0.032	0.250	0.017	0.004	0.397

(d) Quiet-Fault Point Recall

Injected fault	Unif.	Var.	MAD	`lowvar`	`dev_maxz`
Flatline	0.021	0.043	0.058	0.182	0.087
Flatline (mid-var.)	0.024	0.048	0.068	0.317	0.087
Stuck sensor	0.021	0.043	0.058	0.182	0.087
Dropout	0.020	0.043	0.058	0.182	0.087
Low-amp. step	0.030	0.049	0.062	0.182	0.087
$+5\sigma$ spike	0.023	0.042	0.059	0.182	0.108
Mean nom. FPR	0.019	0.069	0.055	0.149	0.141

(e) Controlled Mechanism-Discrimination Study (19 Non-GHL Streams)

	Tail/Drift		Held FPR		Point-F1		PR-AUC		Causal Event Recall			
Arm	γ_2	D_{KS}	Caus.	Buf.	Caus.	Buf.	Caus.	Buf.	SPOT	q^*=0.05	f^*=0.01	f^*=0.05
`TS-JEPA`	1.33	0.214	0.024	0.093	0.161	0.206	0.278	0.331	0.487	0.730	0.395	0.584
`OpEntropy`	0.94	0.209	0.028	0.101	0.153	0.233	0.280	**0.351**	0.517	0.752	0.371	0.573
`JEPA-CtxRecon`	2.59	0.203	0.017	0.108	0.112	0.188	0.235	0.256	0.542	0.795	**0.431**	0.573
`JEPA-NoEMA`	1.16	**0.167**	**0.009**	0.098	0.146	**0.235**	0.273	0.328	0.503	**0.816**	0.398	0.600
`TS-JEPA-NoGate`	0.91	0.214	0.018	0.077	0.156	0.209	0.272	0.324	0.538	0.728	0.418	0.599
`PCA-LinPred-LN`	0.95	0.233	0.021	0.186	0.017	0.104	0.210	0.244	0.452	0.665	0.377	**0.624**
`TCN-ObsPred-LN`	0.91	0.376	0.117	0.193	0.084	0.126	0.235	0.257	0.421	0.665	0.329	0.541
`IsoForest`	**0.70**	0.373	0.039	**0.056**	0.041	0.041	0.209	0.222	0.149	0.631	0.317	0.443
`TCN-ObsPred`	6.10	0.463	0.247	0.313	0.140	0.145	0.272	0.310	0.546	0.629	0.171	0.383
`TCN-ObsRecon`	6.24	0.445	0.245	0.304	0.134	0.135	0.261	0.301	0.528	0.619	0.163	0.379
`TimesNet`	5.54	0.438	0.243	0.300	**0.209**	0.197	**0.293**	0.329	**0.588**	0.618	0.217	0.407
`TranAD`	5.64	0.459	0.222	0.307	0.140	0.130	0.248	0.287	0.533	0.602	0.149	0.343

(0.133 vs. 0.177), while on Genesis, Metro, and SWAN reconstructors record zero FPR with zero F1 due to inactivity.

Point-F1, Event Recall, and Ranking Uncertainty: False-alarm suppression does not establish universal detection superiority. On the 19 non-GHL streams, buffered Point-F1 (95% CI) and event recall (95% CI) are 0.206 $[0.113, 0.319]$ and 0.477 $[0.331, 0.629]$ for TS-JEPA, and 0.233 $[0.131, 0.355]$ and 0.550 $[0.398, 0.700]$ for OpEntropy, versus 0.197 $[0.098, 0.315]$ and 0.583 $[0.383, 0.781]$ for TimesNet, and 0.130 $[0.053, 0.232]$ and 0.490 $[0.295, 0.679]$ for TranAD. Across all 33 streams paired differences remain inconclusive: vs. TimesNet $+0.012$ F1 (CI $[-0.053, +0.075]$, p=0.54), -0.000 PR-AUC (p=0.39); vs. TranAD $+0.051$ F1 (p=0.88), $+0.023$ PR-AUC (p=0.16). Between-stream SDs match the means (SMAP 0.379 ± 0.361 for TS-JEPA vs. 0.289 ± 0.334 for TimesNet). Only SMAP exceeds the constant-alarm reference $\text{F1}_{\text{all-pos}}$=0.236 (0.379 and 0.396); MSL, Daphnet, and SWAN fall below it, while PR-AUC leads over prevalence on SMAP (0.504/0.525 at π=0.150) and MSL (0.434/0.422 at π=0.255).

Domain Boundaries: Latent scoring benefits multi-channel spacecraft and kinematic streams with localized transients, whereas quiet flatlines are filtered with ambient noise (Table I(d)) and continuous process ramps (GHL, Metro) defeat all tested detectors (0.000–0.022 F1).

B. Evaluating Physical Inductive Priors: Hypothesis Testing and Null Results

What does the Operator-Entropy formulation contribute beyond unconstrained `TS-JEPA`? While isolating the spectral regularizer on the fixed Koopman backbone (λ_{ent}=0 control) confirms strict TOST equivalence within $\Delta = \pm 0.03$ Point-F1 (-0.004, 90% CI $[-0.011, +0.003]$, $p_{\text{TOST}} = 0.00074$, Section VI-A), the OpEntropy transition architecture (linear Koopman operator $\mathbf{K}$, detached Lyapunov alignment, and pre-LayerNorm relative radial energy scoring) shortens buffered detection delay from 110.3 steps (95% CI $[70.3, 151.9]$) in TS-JEPA to 79.5 steps ($[47.9, 113.2]$). Across all 33 streams, cross-architecture Point-F1 differences against TS-JEPA satisfy TOST(± 0.05) for OpEntropy ($+0.016$, 90% CI $[-0.009, +0.045]$, two-sided p=0.23; $p_{\text{lower}} = 8.8 \times 10^{-6}, p_{\text{upper}} = 0.0019, p_{\text{TOST}} = 0.0019$) and Reynolds-Stress ($-0.008$, 90% CI $[-0.031, +0.016]$, $p_{\text{TOST}} = 0.0017$), whereas on the 19-stream non-GHL subset the $+0.028$ OpEntropy difference (90% CI $[-0.018, +0.074]$, $p_{\text{TOST}} = 0.11$) is inconclusive. Potential-Flow degrades Point-F1 (-0.081, p=0.033 unadjusted). Physical regularizers therefore act as structural conditioning constraints rather than detection drivers.

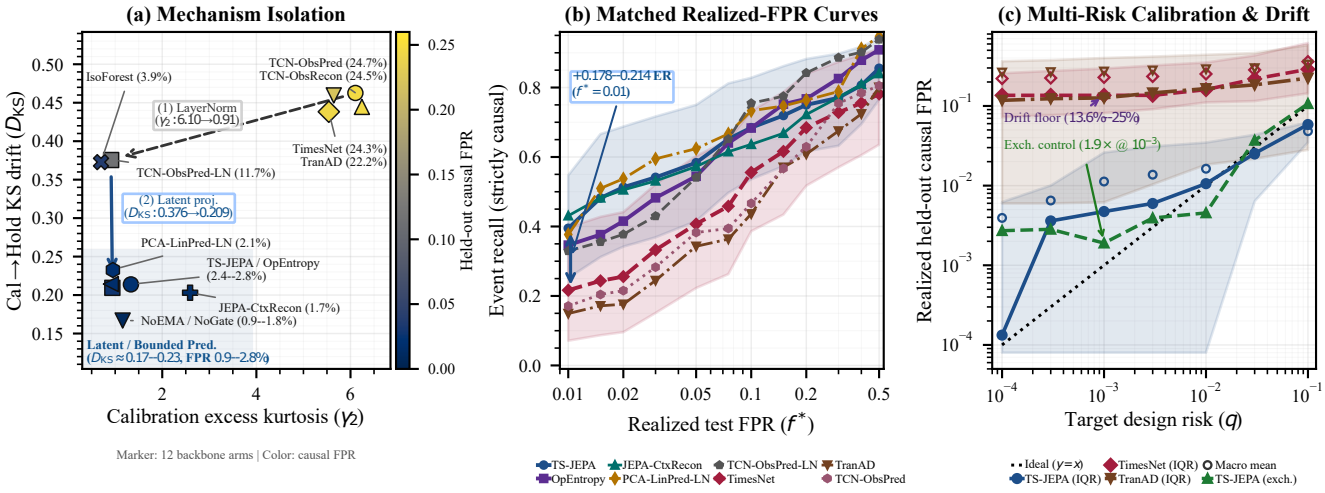


Fig. 2. Residual conditioning, operating curves, and calibration reliability across 19 non-GHL streams (five seeds, strictly causal trailing inference). **(a)** Calibration excess kurtosis (γ_2) vs. calibration-to-holdout KS drift (D_{KS}), colored by held-out causal FPR at SPOT $q{=}10^{-3}$. **(b)** Causal event recall across matched realized test-FPR budgets $f^* \in [0.01, 0.50]$ (95% stream-bootstrap CI bands). **(c)** Realized causal FPR (median lines, IQR bands, and hollow macro means) across design risks $q \in [10^{-4}, 10^{-1}]$ with an exchangeable point-shuffle control.

TABLE II
HELD-OUT CALIBRATION BENCHMARK (SPOT, $u{=}p_{98}$, $q{=}10^{-3}$): (A) RETROSPECTIVE BUFFERED ($S{=}64$) AND (B) STRICTLY CAUSAL TRAILING ATTRIBUTION ACROSS 19 NON-GHL AND 14 GHL STREAMS, (C) PER-DATASET BREAKDOWN (MEAN ± SD), AND (D) 130-STREAM SINGLE-SEED CROSS-CHECK. BEST IN **BOLD**, SECOND UNDERLINED.

	(a) Buffered, non-GHL (19 streams)					(b) Causal, non-GHL			GHL buffered (14)		Summary
Model	FPR ↓	F1 ↑	PR-AUC ↑	ER ↑	Δt ↓	FPR ↓	F1 ↑	ER ↑	FPR ↓	F1 ↑	DS-Macro F1 ↑
Latent-space predictive architectures (JEPAs)											
TS-JEPA (unconstrained)	0.093	0.206	0.331	0.477	110.3	0.024	0.161	0.487	0.004	0.000	0.115
Operator-Entropy JEPA	0.101	**0.233**	**0.351**	0.550	79.5	0.028	0.153	0.517	0.012	0.001	**0.125**
Reynolds-Stress JEPA	0.074	0.190	0.313	0.457	160.9	0.011	0.131	0.491	**0.003**	0.002	0.101
Potential-Flow JEPA	0.087	0.138	0.235	0.449	104.1	**0.008**	0.029	0.459	0.011	0.006	0.089
Observation-space reconstructive baselines											
TimesNet	0.300	0.197	0.329	**0.583**	**48.5**	0.243	**0.209**	**0.588**	0.061	**0.022**	0.100
TranAD	0.307	0.130	0.287	0.490	234.4	0.222	0.140	0.533	0.046	0.020	0.069
Non-neural linear baseline											
Linear AR Forecaster	**0.005**	0.051	0.255	0.432	87.8	–	–	–	**0.002**	0.000	0.038

	(c) Dataset descriptors				Buffered Point-F1 (mean ± SD)				Buffered held-out FPR			
Dataset	n	K	π	$F1_{all\text{-}pos}$	TS-JEPA	OpEnt	TimesNet	TranAD	TS-JEPA	OpEnt	TimesNet	TranAD
Daphnet	6	9	.148	.256	0.145±0.058	0.122±0.061	0.229±0.232	0.144±0.119	0.091	0.059	0.313	0.298
GHL	14	16	.013	.026	0.000±0.000	0.001±0.003	0.022±0.030	0.020±0.025	0.004	0.012	0.061	0.046
Genesis	1	18	.003	.006	0.000±0.000	0.008±0.017	0.000±0.000	0.000±0.000	0.017	0.082	0.000	0.000
MSL	4	55	.255	.381	0.161±0.119	0.323±0.263	0.158±0.146	0.151±0.149	0.095	0.177	0.133	0.241
SMAP	6	25	.150	.236	0.379±0.361	0.396±0.329	0.289±0.334	0.167±0.341	0.135	0.125	0.549	0.513
Metro	1	5	.001	.003	0.000±0.000	0.000±0.000	0.001±0.001	0.000±0.000	0.014	0.026	0.000	0.000
SWAN	1	38	.279	.437	0.119±0.048	0.024±0.031	0.000±0.000	0.000±0.000	0.000	0.000	0.000	0.000

(d) 130-stream cross-check (Point-F1, dataset-macro / stream mean): NCAD 0.058/0.088; DCdetector 0.070/0.097; Anomaly Transformer 0.091/0.082; TimesNet 0.129/0.151; TranAD 0.156/0.152; TS-JEPA 0.160/0.186.

C. Operational Incident Evaluation and Causal-vs-Buffered Trade-offs

Strictly causal results provide the operational evaluation; buffered results are retrospective. Table II(b) reports strictly causal streaming. On 19 non-GHL streams, TS-JEPA reduces held-out nominal FPR to 0.024 (95% CI $[0.007, 0.046]$; OpEntropy 0.028 $[0.007, 0.063]$, Reynolds 0.011 $[0.006, 0.017]$) versus 0.243 $[0.123, 0.375]$ (TimesNet) and 0.222 $[0.119, 0.337]$ (TranAD). Causal Point-F1 is 0.161 $[0.063, 0.287]$ (TS-JEPA) and 0.153 $[0.070, 0.247]$ (OpEntropy) versus 0.209 $[0.109, 0.326]$ (TimesNet), with causal event recall at 0.487 $[0.345, 0.627]$ and 0.517 $[0.367, 0.661]$ vs. 0.588 $[0.394, 0.783]$ (TimesNet). Buffered detection delay is 79.5 $[47.9, 113.2]$ steps for OpEntropy vs. 110.3 $[70.3, 151.9]$ for TS-JEPA, 234.4 $[31.0, 603.1]$ for TranAD, and 48.5 $[20.0, 85.1]$ for TimesNet (causal delay 112.5–124.0 vs. 62.2 steps): without lookahead buffers, latent scoring operates as a conservative filter trading raw recall and onset speed for 8–20× false-alarm suppression. Dynamic SPOT on five incident streams suppresses false alarms up to 21× under drift.

D. Horizon Length, Score-Emission Latency, and False-Alarm Control

At $S{=}64$, scores emit 63 steps after window opening (10.5 min on 10-s telemetry; 63 h on hourly Metro data). Table III(a) retrains models at $S \in \{1, 4, 16, 32, 64\}$ on five streams (three seeds). Shortening horizon preserves false-alarm control: at $S{=}1$ (no emission lag), OpEntropy holds causal FPR at 0.005 (TimesNet 0.025) with event recall 0.729 (TimesNet 0.895). Causal FPR for reconstructors rises

TABLE III
OPERATIONAL LATENCY AND CALIBRATION-QUANTILE TRADE-OFFS: (A) TARGET-HORIZON SWEEP ($S \in \{1, 4, 16, 32, 64\}$, LAG $S-1$, n=5 STREAMS, 3 SEEDS) AND (B) DETECTION AT DEPLOYABLE NOMINAL CALIBRATION QUANTILES q^* (n=19 STREAMS; ER = EVENT RECALL, P = PRECISION, FA = FALSE ALARMS PER 1,000 STEPS).

(a) Target-Horizon and Score-Emission Sweep

		OpEntropy, causal			TS-JEPA, causal		TimesNet, causal		OpEntropy, buf	
S	Lag	FPR	ER	Δt	FPR	ER	FPR	ER	FPR	ER
1	0	0.005	0.729	76.3	0.004	0.524	0.025	0.895	0.005	0.729
4	3	0.007	0.624	179.3	0.006	0.395	0.031	0.910	0.020	0.481
16	15	0.003	0.257	200.3	0.003	0.257	0.061	0.700	0.043	0.391
32	31	0.011	0.219	397.9	0.003	0.276	0.098	0.657	0.057	0.376
64	63	0.020	0.410	209.8	0.004	0.200	0.213	0.614	0.099	0.367

(b) Detection at Matched Nominal Budgets q^*

		TS-JEPA			OpEntropy			TimesNet			TranAD		
Regime	q^*	ER	P	FA	ER	P	FA	ER	P	FA	ER	P	FA
Causal	0.001	0.493	0.285	88.1	0.556	0.228	93.7	0.590	0.219	222.4	0.533	0.156	209.6
Causal	0.050	0.730	0.251	177.3	0.752	0.259	202.8	0.618	0.221	262.3	0.602	0.159	269.5
Buffered	0.001	0.487	0.236	219.4	0.559	0.242	247.3	0.583	0.194	259.2	0.516	0.130	262.6
Buffered	0.050	0.582	0.233	283.7	0.654	0.268	317.6	0.612	0.196	287.3	0.596	0.145	295.3

steeply with horizon ($0.025 \rightarrow 0.213$), while latent models remain at 0.003–0.020. Buffered attribution inflates FPR with S ($0.005 \rightarrow 0.099$ for OpEntropy). Where rapid response matters, S=1 to 4 is the natural operating regime.

E. Deployable Calibration Quantiles vs. Matched Realized Test-FPR Operating Points

We strictly separate two operating evaluations: (i) *deployable calibration quantiles* ($q^* \in [10^{-4}, 0.9]$, Table III(b)), selected using nominal calibration data alone, and (ii) *post-hoc matched realized test-FPR curves* ($f^* \in \{0.01, 0.05, 0.10, 0.20\}$, Fig. 2(b), Table I(e)), which set oracle thresholds on test nominal scores to compare intrinsic receiver operating separation at equalized alarm rates. Under deployable calibration quantile q^*=0.05 causally, TS-JEPA achieves event recall 0.730 at 177.3 false alarms per 1,000 nominal steps (precision 0.251) and OpEntropy reaches 0.752 at 202.8 alarms (0.259) versus 0.618 at 262.3 alarms (0.221) for TimesNet; buffered at q^*=0.001, TS-JEPA attains recall 0.487 (OpEntropy 0.559) vs. 0.583 for TimesNet at higher precision (0.236–0.242 vs. 0.194). Because anomalies span hundreds of steps (mean 402 in Daphnet, 1,102 in SMAP), alarming at 30% FPR intersects $> 95\%$ of segments by chance ($1 - (1 - \text{FPR})^L$).

Under post-hoc matched realized test FPR ($f^* = 0.01$ strictly causal), TS-JEPA achieves event recall 0.395 (latent models 0.371–0.431, reaching 0.431 for `JEPA-CtxRecon`) versus 0.217 for TimesNet and 0.149 for TranAD. At relaxed realized budgets ($f^* \geq 0.10$ causal and across smear mappings), event recall differences narrow or favor reconstructors (over $f^* \in [0.10, 0.50]$, causal TS-JEPA vs. TimesNet is $+0.061$, 95% CI $[-0.153, +0.329]$, $p = 0.625$). Latent scoring thus improves sensitivity specifically at stringent low false-alarm budgets ($f^* \leq 0.01$), while achieving those budgets prospectively without test labels remains bounded by calibration drift.

F. Automated Threshold Calibration Realities on Non-Stationary Telemetry

Does static extreme-value thresholding achieve its nominal design risk q=10^{-3} (0.1%) out of sample? It does not. Across all 33 streams (Table IV(a), where Overshoot $> q$ is the share of cells whose 95% CI lower bound exceeds q), standard SPOT achieves cell-weighted (stream-macro) causal FPR of 1.71% (1.51%) for TS-JEPA and 2.12% (1.95%) for OpEntropy, and buffered FPR of 6.40% (5.54%) and 7.25% (6.38%),

TABLE IV
THRESHOLD CALIBRATOR COMPARISON AT DESIGN RISK q=10^{-3}: (A) STATIC RULES ACROSS 33 STREAMS (CELL-WEIGHTED MEANS, STREAM-MACRO IN PARENTHESES) AND (B) CONTROLLED-DRIFT STRESS TEST ACROSS 16 STREAMS.

Model Architecture	Calibrator Rule	Causal FPR	Overshoot $> q$	Buffered FPR	Point-F1 (buf)
TS-JEPA (Control)	Standard SPOT ($q = 10^{-3}$)	0.0171 (0.0151)	10%	0.0640 (0.0554)	0.1386
	SPOT + Kurtosis ($\beta = 0.167$)	0.0185 (0.0159)	10%	0.1630 (0.1397)	0.1498
	Empirical 99.5th Percentile	0.0254 (0.0229)	21%	0.0756 (0.0659)	0.1399
	Validation Maximum	0.0160 (0.0140)	7%	0.0658 (0.0568)	0.1385
Operator-Entropy	Standard SPOT ($q = 10^{-3}$)	0.0212 (0.0195)	14%	0.0725 (0.0638)	0.1573
	SPOT + Kurtosis ($\beta = 0.167$)	0.0209 (0.0188)	10%	0.1346 (0.1165)	0.1610
	Empirical 99.5th Percentile	0.0298 (0.0281)	27%	0.0882 (0.0783)	0.1633
	Validation Maximum	0.0187 (0.0170)	10%	0.0727 (0.0637)	0.1566
TimesNet	Standard SPOT ($q = 10^{-3}$)	0.1795 (0.1611)	49%	0.2217 (0.1985)	0.1395
	SPOT + Kurtosis ($\beta = 0.167$)	0.1843 (0.1623)	46%	0.2281 (0.2011)	0.1352
	Empirical 99.5th Percentile	0.1914 (0.1752)	51%	0.2338 (0.2125)	0.1413
	Validation Maximum	0.1791 (0.1599)	48%	0.2263 (0.2032)	0.1410
TranAD	Standard SPOT ($q = 10^{-3}$)	0.1601 (0.1423)	46%	0.2217 (0.1963)	0.0936
	SPOT + Kurtosis ($\beta = 0.167$)	0.1722 (0.1513)	43%	0.2453 (0.2151)	0.0940
	Empirical 99.5th Percentile	0.1712 (0.1544)	51%	0.2343 (0.2093)	0.0962
	Validation Maximum	0.1580 (0.1392)	44%	0.2270 (0.2009)	0.0952

(b) Injected condition	Static SPOT	Static empirical	Rolling empirical	Rolling SPOT
Stationary (no injection)	0.061	0.162	0.016	0.047
Mean shift $+1.0\sigma$	0.087	0.216	0.011	0.043
Regime change $+1\sigma$	0.073	0.187	0.015	0.070
Variance scale 2.0	0.082	0.195	0.016	0.047
Autocorrelation 0.9	0.113	0.232	0.024	0.055
Continuous mean ramp	0.075	0.186	0.326	0.385

overshooting the 0.1% target by 15–73× due to deployment shift, though substantially less than TimesNet (17.95% causal, 22.17% buffered) and TranAD (16.01%, 22.17%). Three diagnostic findings explain this behavior:

1) **Multi-risk breakdown and quantile/length sensitivity:** Across $q \in \{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}\}$ (Fig. 2(c)), static thresholding overshoots nominal risk for all models: TS-JEPA median causal FPR is 0.0001, 0.0047, 0.0106, and 0.0585 vs. 0.1360, 0.1360, 0.1603, and 0.2889 for TimesNet. On the 19 non-GHL streams at u=p_{98}, median calibration exceedance count is N_u=28 (IQR 20–31) causally and 21 (16–32) buffered, with 0% numerical fallback and median bootstrap threshold uncertainty $\text{SE}(\tau)/\tau$ = 2.6%–3.5% causal (0.8%–1.1% buffered). Causally, Grimshaw GPD fits hit the lower shape boundary (ξ= − 0.90) in 4.4% of TS-JEPA and OpEntropy cells (median ξ= − 0.27, σ=0.11) vs. 34.4% (TimesNet) and 54.4% (TranAD); buffered 64-step smoothing raises boundary hits to 26.3%–31.6%. Varying $u \in \{p_{95}, p_{98}, p_{99}\}$ leaves causal FPR invariant (0.0241–0.0242 for TS-JEPA; 0.0278 for OpEntropy), whereas halving calibration length (50% vs. 100% N_{cal}) lowers median N_u to 20 and raises causal FPR from 0.0242 to 0.0372 (TS-JEPA) and 0.0278 to 0.0413 (OpEntropy).
2) **Mechanism of the kurtosis-multiplier failure:** Code verification confirms $\tau_{\text{eff}} = \tau[1 + \frac{1}{6}\tanh(\gamma_2)]$ to $< 10^{-15}$. On leptokurtic buffered cells ($\gamma_2 \geq 0$, 68.1% of TS-JEPA cells), $\tau_{\text{eff}} \geq \tau$ lowers buffered FPR from 0.0449 to 0.0111 (OpEntropy 0.0634 $\rightarrow$ 0.0386). However, 64-

step smoothing renders 31.9% of buffered cells platykurtic ($\gamma_2 < 0$) with tight dispersion ($\sigma_{\text{cal}}/\mu_{\text{cal}} \approx 0.02$–$0.05$): scaling total τ toward zero pulls τ_{eff} below $p_{99.5}$ in 28.4% of cells, inflating FPR on $\gamma_2 < 0$ cells from 0.1048 to 0.4870 (0.0640 → 0.1630 overall). Scaling strictly the tail excess above u, $\tau_{\text{exc}} = u + (\tau - u)[1 + \frac{1}{6}\tanh(\gamma_2)] \geq u$, prevents threshold collapse below u and holds 19-stream buffered FPR to 0.0919 vs. 0.0929 unadjusted for TS-JEPA (0.1008 vs. 0.1011 for OpEntropy; 0.0240 vs. 0.0242 causal).

3) **Three-stage permutation decomposition of drift vs. temporal dependence:** To separate inter-block shift from autocorrelation, we decompose total causal excess overshoot ($\Delta\text{FPR}_{\text{tot}} = \text{FPR}_{\text{chrono}} - q$ at $q{=}10^{-3}$) across chronological ($\text{FPR}_{\text{chrono}}$), contiguous $L{=}64$ block permutation ($\text{FPR}_{\text{block}}$, equalizing block distributions while preserving intra-block dependence), and pointwise i.i.d. permutation ($\text{FPR}_{\text{point}}$). For TS-JEPA (0.0121 → 0.0058 → 0.0019), inter-block shift ($\text{FPR}_{\text{chrono}} - \text{FPR}_{\text{block}}$) accounts for +0.0063 (95% CI $[-0.0054, +0.0194]$, 56.6% of excess), serial dependence ($\text{FPR}_{\text{block}} - \text{FPR}_{\text{point}}$) for +0.0039 ($[+0.0001, +0.0103]$, 35.3%), and i.i.d. tail error ($\text{FPR}_{\text{point}} - q$) for +0.0009 ($[-0.0006, +0.0027]$, 8.1%). For OpEntropy (0.0185 → 0.0073 → 0.0022), inter-block shift accounts for 64.1% (+0.0112 $[-0.0015, +0.0336]$), serial dependence for 29.1% (+0.0051 $[-0.0005, +0.0146]$), and tail error for 6.9%. For TimesNet (0.0872 → 0.0204 → 0.0037) and TranAD (0.0741 → 0.0067 → 0.0032), inter-block shift accounts for 77.5% and 92.3% of excess, and serial dependence for 19.4% and 4.7%. Across $N{=}1{,}080$ cells, KS drift correlates with realized FPR ($\rho{=}+0.377$, CI $[+0.314, +0.433]$). In Table IV(b), rolling calibrators hold risk to 1%–7% under discrete shifts but fail under continuous ramps (0.33–0.39 FPR).

VIII. Conclusion and Future Work

We formulated Operator-Entropy JEPA for multi-sensor anomaly detection and evaluated relative false-alarm control, discriminative detection, and extreme-value calibration across 33 telemetry streams under a strict held-out protocol:

1) **Mechanism Isolation and Regularizer Equivalence Bounds:** Contrasting 12 controlled backbones decouples score compression from representation learning: LayerNorm bottlenecks mechanically bound kurtosis (γ_2 : 6.10 → 0.91) and halve FPR, while learned latent projection halves calibration-to-holdout distribution drift (0.376 → 0.209–0.214) and suppresses causal FPR to 0.009–0.028. On a fixed Koopman backbone, the spectral regularizer confirms formal TOST equivalence within $\Delta = \pm 0.03$ Point-F1 ($p_{\text{TOST}} = 0.00074$; Reynolds-Stress within ± 0.05, $p_{\text{TOST}} = 0.0019$), while the OpEntropy transition architecture reduces buffered detection delay from 110.3 to 79.5 steps (127.0 → 58.2 causal) with cross-architecture Point-F1 inconclusive on the 19-stream subset ($p_{\text{TOST}} = 0.11$) and bounded within ± 0.05 across all 33 streams ($p_{\text{TOST}} = 0.0019$).
2) **Relative False-Alarm Control vs. Detection Sensitivity:** Latent predictive scoring suppresses nominal false alarms under static calibration three- to four-fold buffered (7.4%–10.1% vs. 30.0%–30.7%) and 8–20× causally (1.1%–2.8% vs. 22.2%–24.3%) against TimesNet and TranAD across leave-one-dataset-out folds. At post-hoc matched realized test FPR ($f^* = 0.01$ causal), latent predictors reach event recall 0.371–0.431 (0.395 for TS-JEPA) versus 0.217 for TimesNet and 0.149 for TranAD, whereas at $f^* \geq 0.10$ and across aggregate Point-F1 and PR-AUC, detection remains comparable to reconstructors.
3) **Calibration Breakdown and Drift vs. Dependence Decomposition:** Static extreme-value thresholding does not achieve its nominal $q{=}10^{-3}$ design target out of sample (1.5%–2.0% stream-macro causal and 5.5%–6.4% buffered FPR across 33 streams for JEPAs vs. 14%–20% for reconstructors). Under our three-stage permutation protocol, inter-block shift accounts for 56.6%–64.1% of excess causal overshoot for JEPAs (77.5%–92.3% for reconstructors) and intra-block serial dependence accounts for 29.1%–35.3% (4.7%–19.4% for reconstructors). Rolling thresholding tracks discrete shifts but fails under continuous drift ramps.
4) **Latency Limits and Fault Boundaries:** Predictive scoring at $S{=}64$ incurs a 63-step emission lag (10.5 min on 10-s telemetry). While zero-lag monitoring ($S{=}1$) preserves false-alarm control (0.005 vs. 0.025), quiet flatlines remain undetected without high-alarm auxiliary variance monitors.

Open items include adaptive thresholding tracking within-block drift and low-latency architectures for quiet fault modes.

References

[1] F. Meng, B. Yuan, and F. Gao, "Spacecraft telemetry anomaly detection based on deep learning: A review and benchmark," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 56, no. 5, pp. 3863–3878, 2020.

[2] R. Chalapathy and S. Chawla, "Deep learning for anomaly detection: A survey," *arXiv preprint arXiv:1901.03407*, 2019.

[3] K. Hundman, V. Constantinou, C. Laporte, I. Colwell, and T. Soderstrom, "Detecting spacecraft anomalies using lstms and nonparametric dynamic thresholding," *Proceedings of the 24th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, pp. 387–395, 2018.

[4] J. Xu, H. Wu, J. Wang, and M. Long, "Anomaly transformer: Time series anomaly detection with association discrepancy," in *International Conference on Learning Representations (ICLR)*, 2022.

[5] H. Wu, T. Hu, Y. Liu, H. Zhou, J. Wang, and M. Long, "Timesnet: Temporal 2d-variation modeling for general time series analysis," in *International Conference on Learning Representations (ICLR)*, 2023.

[6] S. Tuli, G. Casale, and N. R. Jennings, "Tranad: Deep transformer networks for anomaly detection in multivariate time series data," *Proceedings of the VLDB Endowment*, vol. 15, no. 6, pp. 1201–1214, 2022.

[7] A. Siffer, P.-A. Fouque, A. Termier, and C. Largouet, "Anomaly detection in streams with extreme value theory," in *Proceedings of the 23rd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2017, pp. 1067–1075.

[8] C. U. Carmona, F.-X. Aubet, V. Flunkert, and J. Gasthaus, "Neural contextual anomaly detection for time series," in *International Conference on Learning Representations (ICLR)*, 2022.

[9] Y. LeCun, "A path towards autonomous machine intelligence," *OpenReview Preprint*, 2022.

[10] S. L. Brunton, J. L. Proctor, and J. N. Kutz, "Discovering governing equations from data by sparse identification of nonlinear dynamical systems," *Proceedings of the National Academy of Sciences*, vol. 113, no. 15, pp. 3932–3937, 2016.

[11] R. T. Chen, Y. Rubanova, J. Bettencourt, and D. K. Duvenaud, "Neural ordinary differential equations," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 31, 2018, pp. 6571–6583.

[12] S. L. Brunton, B. W. Brunton, J. L. Proctor, and J. N. Kutz, "Koopman invariant subspaces and finite-dimensional approximations of nonlinear

dynamical systems," *Journal of Nonlinear Science*, vol. 26, pp. 1131–1162, 2016.
[13] B. Lusch, J. N. Kutz, and S. L. Brunton, "Deep learning for universal linear embeddings of nonlinear dynamics," *Nature Communications*, vol. 9, no. 1, p. 4950, 2018.
[14] Y. Liu, C. Li, J. Wang, and M. Long, "Koopa: Learning non-stationary time series dynamics with Koopman predictors," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 36, 2023, pp. 13 480–13 492.
[15] Y. Yang, C. Zhang, T. Zhou, Q. Gao, Q. Wen, and Y. Liang, "Dcdetector: Dual attention contrastive representation learning for time series anomaly detection," in *Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 2023, pp. 3033–3045.
[16] M. Assran, Q. Duval, I. Misra, P. Bojanowski, P. Vincent, M. Rabbat, Y. LeCun, and N. Ballas, "Self-supervised learning from images with a joint-embedding predictive architecture," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023, pp. 15 619–15 629.
[17] A. Bardes, Q. Duval, C. Feichtenhofer, I. Laptev, D. Haziza, V. Khalidov, N. Ballas, and Y. LeCun, "V-jepa: Video joint-embedding predictive architecture," *arXiv preprint arXiv:2404.08471*, 2024.
[18] J. Zbontar, L. Jing, I. Misra, Y. LeCun, and S. Deny, "Barlow twins: Self-supervised learning via redundancy reduction," in *International Conference on Machine Learning (ICML)*, 2021, pp. 12 310–12 320.
[19] A. Ermolov, A. Siarohin, E. Sangineto, and N. Sebe, "Whitening for self-supervised representation learning," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 34, 2021, pp. 8315–8327.
[20] A. Bardes, J. Ponce, and Y. LeCun, "Vicreg: Variance-invariance-covariance regularization for self-supervised learning," in *International Conference on Learning Representations (ICLR)*, 2022.
[21] Z. Yue, Y. Wang, J. Duan, T. Yang, C. Huang, Y. Tong, and B. Xu, "Ts2vec: Towards universal representation of time series," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 36, no. 8, 2022, pp. 8980–8987.
[22] M. Raissi, P. Perdikaris, and G. E. Karniadakis, "Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations," *Journal of Computational Physics*, vol. 378, pp. 686–707, 2019.
[23] S. Greydanus, M. Dzamba, and J. Yosinski, "Hamiltonian neural networks," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 32, 2019, pp. 15 379–15 389.
[24] M. Cranmer, S. Greydanus, S. Hoyer, P. Battaglia, D. Spergel, and S. Ho, "Lagrangian neural networks," *arXiv preprint arXiv:2003.04630*, 2020.
[25] S. Wang, Y. Teng, and P. Perdikaris, "Understanding and mitigating gradient pathologies in physics-informed neural networks," *SIAM Journal on Scientific Computing*, vol. 43, no. 5, pp. A3055–A3081, 2021.
[26] M. Bächlin, M. Plotnik, D. Roggen, I. Maidan, J. M. Hausdorff, N. Giladi, and G. Tröster, "Wearable assistant for parkinson's disease patients with the freezing of gait symptom," *IEEE Transactions on Information Technology in Biomedicine*, vol. 14, no. 2, pp. 436–446, 2010.
[27] A. Asuncion and D. Newman, "UCI Machine Learning Repository: CalIt2 Building People Counts Data Set," https://archive.ics.uci.edu/dataset/179/calit2+building+people+counts, 2007.
[28] S. Schmidl, P. Wenig, and T. Papenbrock, "Anomaly detection in time series: a comprehensive evaluation," *Proceedings of the VLDB Endowment*, vol. 15, no. 9, pp. 1779–1797, 2022.
[29] Y. Zhou, C. Brif, and I. Lourentzou, "mTSBench: A Comprehensive Benchmark for Multi-Domain Time Series Anomaly Detection," *Transactions on Machine Learning Research (TMLR)*, 2026, arXiv:2506.21550.
[30] R. A. Angryk, P. C. Martens, B. Aydin, D. Kempton, S. S. Mahajan, S. Basodi, A. Ahmadzadeh, S. Boubrahimi, S. M. Hamdi, M. A. Schuh *et al.*, "Multivariate time series dataset for space weather data analytics," *Scientific Data*, vol. 7, no. 1, p. 227, 2020.
[31] L. M. Candanedo and V. Feldheim, "Accurate occupancy detection of an office room from light, temperature, humidity and co2 measurements using statistical learning models," *Energy and Buildings*, vol. 112, pp. 28–39, 2016.
[32] R. Wu and E. J. Keogh, "Current time series anomaly detection benchmarks are flawed and are creating the illusion of progress," *IEEE Transactions on Knowledge and Data Engineering*, vol. 35, no. 3, pp. 2421–2429, 2023.
[33] B. O. Koopman, "Hamiltonian systems and transformations in Hilbert space," *Proceedings of the National Academy of Sciences*, vol. 17, no. 5, pp. 315–318, 1931.
[34] I. Mezić, "Spectral properties of dynamical systems, model reduction and decompositions," *Nonlinear Dynamics*, vol. 41, no. 1–3, pp. 309–325, 2005.
[35] J.-B. Grill, F. Strub, F. Altché, C. Tallec, P. Richemond, E. Buchatskaya, C. Doersch, B. Avila Pires, Z. Guo, M. Gheshlaghi Azar *et al.*, "Bootstrap your own latent-a new approach to self-supervised learning," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, 2020, pp. 21 271–21 284.
[36] S. Kim, K. Choi, J. Choi, B. Lee, and S. Yoon, "Towards a rigorous evaluation of time-series anomaly detection," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 36, no. 7, 2022, pp. 7194–7201.